

Low Emotional Engagement Detection using Pupil Tracking and Physiological Signals: A Multimodal Approach

Abstract

Emotional engagement significantly influences user attention, motivation, and cognitive performance. This work targets the detection of *low emotional engagement* operationalized as moments in the low-valence, low-arousal (LV–LA) quadrant of the circumplex model, (Russell, 1980) using synchronised pupil-tracking and physiological signals. Unlike prior studies relying on post-hoc surveys or single-modality sensing (Matsuda et al., 2022; Whitehill et al., 2018; AlZoubi et al., 2020), our approach fuses high-frequency ocular metrics from a Pupil Labs eye tracker (250 Hz) with galvanic skin response (GSR) and heart rate (HR) data from wearable sensors (10 Hz), alongside continuous in-situ valence–arousal annotations (Banik et al., 2024; Sharma et al., 2019; Sawalha et al., 2017).

Our experimental design employs curated, emotionally diverse video stimuli selected based on the Circumplex Model of Affect, which organises emotions along the dimensions of valence and arousal. Two clips are chosen from each of the four quadrants of this model, ensuring coverage of a wide affective space. Each clip is preceded by a neutral blue screen to reset the participant’s affective state. Participant-specific median thresholds on valence and arousal ratings collected continuously during viewing are used to derive binary engagement labels, where the low-valence, low-arousal (LV–LA) quadrant represents low engagement and all other quadrants represent high engagement. Data are segmented into 5-second windows, with extracted features including pupil diameter, gaze confidence, fixation metrics, GSR peaks, and HR variability.

We propose a multimodal temporal modelling framework and plan to benchmark it against classical models such as Random Forest (RF), XGBoost, and LightGBM, as well as deep temporal architectures like CNN–LSTM (Baltrušaitis et al., 2019). We hypothesise that multimodal fusion of ocular and physiologi-

cal features will outperform unimodal baselines in binary LV–LA classification, with tree-based ensembles expected to provide strong interpretability. We further anticipate that key predictors will include ocular metrics like pupil diameter, gaze confidence and physiological measures such as GSR peaks, HR variability, reflecting both cognitive load and autonomic arousal.

This work advances real-time, affect-grounded engagement detection and has strong potential for adaptive learning and media systems that dynamically respond to engagement dips.

1 Introduction

Emotional engagement plays a pivotal role in sustaining attention, motivation, and information processing (Calvo and D’Mello, 2010). We define *low emotional engagement* as moments in the LV–LA quadrant, linked to boredom, apathy, and reduced cognitive performance (Posner et al., 2005). Detecting these moments in real time enables adaptive systems that adjust content or delivery to maintain engagement.

Despite advances in HCI and affective computing, prior engagement detection work often revolves around: (i) static or domain-specific stimuli, (ii) post-hoc self-reports instead of continuous labelling (Banik et al., 2024; Sharma et al., 2019; Sawalha et al., 2017), (iii) limited integration of physiological and ocular metrics (Matsuda et al., 2022; Wang et al., 2014; Kizilcec et al., 2023; Kawahara et al., 2024; Prajapati et al., 2024), and (iv) small, homogeneous participant samples. Our work addresses these gaps by introducing a multimodal, ecologically valid, real-time engagement detection framework.

Contributions:

1. Operationalisation of LV–LA as low engagement using continuous, participant-personalised valence–arousal annotations (Banik et al., 2024).

2. Creation of a multimodal dataset combining synchronized pupil-tracking, GSR, and HR data with ecologically diverse video stimuli.
3. Implementation of a timestamp-aligned processing pipeline for heterogeneous sensor streams.
4. Comparative evaluation of classical and deep-learning models, with evidence that multimodal fusion improves robustness.

2 Problem Definition

We cast low-emotional-engagement detection as real-time binary sequence labelling grounded in the Circumplex Model of Affect (Russell, 1980). Let $S^{\text{eye}}(t)$ and $S^{\text{phys}}(t)$ denote synchronized ocular (pupil/gaze/fixations) and physiological (GSR/HR) streams, and let $A(t)$ be the continuous self-annotation of valence $V(t)$ and arousal $R(t)$. For participant s , we derive a label $y_t \in \{0, 1\}$ using subject-specific thresholds θ_v^s, θ_r^s :

$$y_t = 1 [V(t) < \theta_v^s \wedge R(t) < \theta_r^s]$$

where $y_t=1$ denotes *low engagement* (LV-LA quadrant) and $y_t=0$ otherwise. Given a causal window $X_{t:t+W}$ constructed from S^{eye} and S^{phys} , the goal is to learn a mapping $f : X_{t:t+W} \rightarrow y_t$ that maximizes detection performance (with emphasis on F1 for the positive class and ROC-AUC) under: (i) real-time constraints (end-to-end latency $\leq \tau$), (ii) robustness to dropouts/missing samples, and (iii) cross-subject generalisation. Evaluation follows subject-independent (leave-one-subject-out) and subject-dependent protocols. Details on sampling rates, featureization, and fusion architectures appear in §??.

3 Methodology

3.1 Data Acquisition

Our study employs the circumplex model of affect (Russell, 1980) to structure stimulus selection, ensuring balanced coverage across all four valence-arousal quadrants. Eight emotionally diverse video clips (two per quadrant) are shown to each participant, with a neutral blue screen inserted before each clip to reset affective state. During viewing, participants:

- Wear a Pupil Labs Pro headset for high-frequency (250 Hz) gaze position and pupil

diameter tracking (Bednarik et al., 2012; Kumar et al., 2025).

- Wear a finger-worn sensor ring to record galvanic skin response (GSR) and heart rate (HR) at 10 Hz (Matthews et al., 2015; Matsuda and et al., 2022).
- Provide continuous self-reported valence-arousal ratings via arrow keys (Banik et al., 2024; Sharma et al., 2019).

3.2 Signal Processing and Synchronisation

All streams (ocular, physiological, and annotation) are synchronised via their timestamps. Engagement labels are derived per participant by applying median thresholds to valence and arousal scores, mapping the low-valence-low-arousal (LV-LA) quadrant to *low engagement* (label 1) and all others to *high engagement* (label 0). Data are segmented into 5-second windows, with extracted features including:

- **Ocular:** mean/variance of pupil diameter, gaze confidence, fixation metrics, blink rate.
- **Physiological:** GSR peaks, skin conductance level, HR variability.

3.3 Modeling and Evaluation

We benchmark multiple modeling approaches:

- **Classical classifiers:** Random Forest (RF), XGBoost, LightGBM.
- **Deep temporal models:** CNN and LSTM like models for multimodal time-series data (Baltrušaitis et al., 2019).

Evaluation uses accuracy, F1-score, and ROC-AUC to compare unimodal and multimodal performance. Feature importance analysis is conducted to assess the relative contribution of ocular and physiological signals.

4 Expected Outcomes

We anticipate that the proposed multimodal fusion approach will significantly outperform unimodal baselines across all evaluation metrics, particularly in scenarios where one modality suffers from transient noise or signal loss. By combining complementary ocular and physiological cues, the framework is expected to capture both attentional focus (via gaze and pupil dynamics) and autonomic

arousal (via GSR and HR), enabling more robust detection of low engagement moments (Prajapati et al., 2024; Rani et al., 2025).

In addition to improved accuracy and F1-score, we expect clearer interpretability of feature contributions through model explainability analyses. Insights from feature importance rankings could guide sensor selection for lightweight, deployable systems in real-world environments. Ultimately, this research is intended to serve as a foundation for adaptive media and e-learning platforms, where real-time engagement monitoring can trigger content adjustments, provide targeted feedback, and personalise user experiences (Kizilcec et al., 2023; Whitehill et al., 2018). Future work will extend these methods to authentic online learning scenarios, incorporating longitudinal data to model user-specific engagement profiles.

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